FEM for Poisson equation:

$$-\Delta u = f$$

v: test function in $H_0^1(\Omega)$

$\downarrow$

weak formulation: $\int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f \cdot v \, dx$

i.e. $\langle \nabla u, \nabla v \rangle = \langle f, v \rangle$

$\quad\quad \parallel$

$\quad a(u, v)$

want to find u s.t. any discrete v in V_h satisfies.

iff all basis functions ϕ_i satisfy.

$$A = (a(\phi_i, \phi_j))_{ij}$$

$u_h = \begin{bmatrix} \vert \end{bmatrix}$ as coordinate coefficients. describe u in V_h

Au_h. $\longrightarrow$ each row e.g. ith row. with u_h.

$$\sum_j u_h^{(j)} a(\phi_i, \phi_j) = \cancel{\int_\Omega \nabla \phi_i \nabla u_h \, dx}$$

$$= a(\phi_i, \sum_j u_h^{(j)} \phi_j) \quad \text{as } a(\cdot, \cdot) \text{ is a bilinear form.}$$

$$= a(\phi_i, u_h)$$

$$= \int_\Omega \nabla \phi_i \nabla u_h \, dx$$

as test with $v = \phi_i$

so corresponding to RHS $\langle f, \phi_i \rangle = \int_\Omega f \phi_i \, dx$.

$\boxed{A u_h = f}$ become linear algebra problem

☆ FEM for Stokes equation:

$$\begin{cases} -\Delta \vec{u} + \nabla p = \vec{f} \\ -\operatorname{div} \vec{u} = g \end{cases} \sim \begin{cases} -\Delta u_1 + \partial_x p = f_1 \\ -\Delta u_2 + \partial_y p = f_2 \\ -\partial_x u_1 - \partial_y u_2 = g \end{cases}$$

$\downarrow$

weak formulation: $(\nabla \vec{u}, \nabla \vec{v}) - (p, \operatorname{div} \vec{v}) = \langle \vec{f}, \vec{v} \rangle$

$-(\operatorname{div} \vec{u}, q) = \langle g, q \rangle = 0$

$\vec{v}$, q are test functions.

simplified.
incompressible

take

in Non-Conforming P_1-P_0 element method.

test function $\vec{v}$, velocity $\vec{u}$ are in P_1^2

(twice, each part in P_1)

test function q, pressure p are in P_0

i.e. piecewise const w.r.t element.

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for $\vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ → represent 2 direction of velocity

u_i : $\underbrace{a_{ij} \int \nabla\phi_i \nabla\phi_j}_{\text{midpoint as basis block A}} + \underbrace{?}_{} = \underbrace{?}_{\text{RHS subroutine.}}$

midpoint as basis

block A

RHS subroutine.

just transpose of left bottom block B
we don't need to worry about it
on this step as it can be done
algebraically after we solving block B.

p : $\underbrace{b_i \int \nabla\phi_i \cdot I}_{\text{block B.}} = \underbrace{?}_{\text{RHS.}}$

RHS.

P_0 basis is const 1.

block B.

☆ Standard Assembling

$$\begin{array}{cc} & \begin{array}{ccc} NE & NE & NT \end{array} \\ \begin{array}{c} NE \\ NE \\ NT \end{array} & \begin{pmatrix} A & 0 & B^T \\ 0 & A & \\ B & & 0 \end{pmatrix} \end{array}$$

global stiffness matrix outline

coordinate representation:

$$\begin{pmatrix} \square \\ \square \\ \square \end{pmatrix} \begin{array}{l} \} NE \\ \} NE \\ \} NT \end{array}$$

NE, NE: non-conforming P_1 use midpoints, so dim = # edges = NE

$NT \to P_0$ piecewise const i.e. pressure on each element is a const so dim = # elements = NT

standard assembling ↑ through local stiffness distribute back to global stiffness

$$\begin{pmatrix} \underset{3\times3}{\tilde{A}} & 0 & \underset{6\times1}{\tilde{B}^T} \\ 0 & \underset{3\times3}{\tilde{A}} & \\ \underset{1\times6}{\tilde{B}} & & 0 \end{pmatrix}$$

local stiffness matrix

confine variational form into single element (triangle)

$$\tilde{A} = 4A_{\text{Poisson}}.$$

as $\tilde{\lambda}_i = 1 - 2\lambda_i \Rightarrow \nabla\tilde{\lambda}_i = -2\nabla\lambda_i$

$$\Rightarrow \underbrace{\int_\tau \nabla\tilde{\lambda}_i \nabla\tilde{\lambda}_j \, d\mu}_{\tilde{A}} = 4 \underbrace{\int_\tau \nabla\lambda_i \nabla\lambda_j \, d\mu}_{A_{\text{Poisson}}}$$

in τ. $q|_\tau \equiv 1$ piecewise const.

$$\int_\tau \partial_x u_1 \cdot 1 \, d\mu = \int_\tau \sum_{i=1}^{3} c_i \partial_x \tilde{\lambda}_i \, d\mu.$$

$$= \sum_{i=1}^{3} c_i \int_\tau \partial_x \tilde{\lambda}_i \, d\mu$$

same. $\int_\tau \partial_y u_2 \cdot 1 d\mu = \sum_{i=1}^{3} d_i \int_\tau \partial_y \tilde{\lambda}_i \, d\mu$

$$A_t(7, \underset{\substack{2\\3}}{1}) = \int_\tau \partial_x \tilde{\lambda}_1 \, d\mu = -2 \int_\tau \partial_x \lambda_1 \, d\mu.$$

$$A_t(7, \underset{\substack{5\\6}}{4}) = \int_\tau \partial_y \tilde{\lambda}_1 \, d\mu = -2 \int_\tau \partial_y \lambda_1 \, d\mu.$$

Recall. in reference simplex. ~~test~~ test func $q \equiv 1$ still 1.

$$\nabla \lambda_1 = (B^{-T} \hat{\nabla} \hat{\lambda}_1)$$

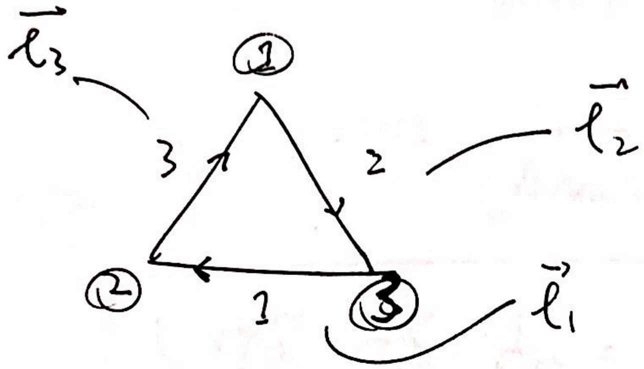
$$\int_\tau \partial_x \lambda_1 \, d\mu = \underbrace{\tfrac{1}{2} |\det(B)|}_{\text{area}} \left. (B^{-T} \underbrace{\hat{\nabla} \hat{\lambda}_1}) \right|_{1^{st} \text{ coord.}}$$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix} \right)$

☆ Vectorized Assembling

rotation matrix $R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \xlongequal{\theta = \frac{\pi}{2}} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

counterclockwise direction



outward $\vec{n_i} = R \cdot \vec{l_i}$

known relation in Poisson FEM

$$\nabla \tilde{\lambda}_i = -2 \cdot \boxed{\nabla \lambda_i} = -2 \cdot \boxed{\frac{1}{\underset{\text{dim.}}{d!}\,\underset{\text{area}}{|\tau|}} \vec{n}_i} \quad \text{inward}$$

(as $\tilde{\lambda}_i = 1 - 2\lambda_i$)

$$= \frac{2}{2!\,|\tau|} \cdot \vec{n}_i$$

now, outward. to cancel with "−" sign

$$= \frac{1}{\text{Area}} \cdot \vec{n}_i \quad \leftarrow \text{outward.}$$

$$= \frac{1}{\text{Area}} \cdot R \cdot \vec{l_i} = \frac{1}{\text{Area}} \begin{bmatrix} -y_{\vec{l_i}} \\ x_{\vec{l_i}} \end{bmatrix}$$

$$\boxed{\partial_x \tilde{\lambda}_i = -\frac{1}{\text{Area}} \cdot y_{\vec{l_i}} \ , \quad \partial_y \tilde{\lambda}_i = \frac{1}{\text{Area}} \cdot x_{\vec{l_i}}}$$

$$\nabla\tilde{\lambda}_i \cdot \nabla\tilde{\lambda}_j = \frac{1}{(\text{Area})^2} \cdot \begin{bmatrix} -y_{\vec{l_i}} \\ x_{\vec{l_i}} \end{bmatrix} \cdot \begin{bmatrix} -y_{\vec{l_j}} \\ x_{\vec{l_j}} \end{bmatrix}$$

$$= \frac{1}{(\text{Area})^2} \cdot \left(y_{\vec{l_i}} \cdot y_{\vec{l_j}} + x_{\vec{l_i}} \cdot x_{\vec{l_j}} \right)$$

$$= \frac{1}{(\text{Area})^2} \cdot \vec{l}_i \cdot \vec{l}_j$$

$$\boxed{\int_\tau \nabla\tilde{\lambda}_i \cdot \nabla\tilde{\lambda}_j \, d\mu = \frac{1}{\text{Area}} \cdot \vec{l}_i \cdot \vec{l}_j} \longrightarrow \text{A block done}$$

comparing with previous Poisson FEM

$$\int_\tau \nabla\lambda_i \cdot \nabla\lambda_j \, d\mu = \frac{1}{(d!)^2 |\tau|} \vec{n}_i \cdot \vec{n}_j$$

$$= \frac{1}{4} \cdot \frac{1}{\text{Area}} \cdot \vec{l}_i \cdot \vec{l}_j$$

$$= \frac{1}{4} \int_\tau \nabla\tilde{\lambda}_i \nabla\tilde{\lambda}_j \, d\mu .$$

✓

$$\cancel{\int_\tau \nabla \tilde{\lambda}_i \cdot 1}$$

$$\boxed{\begin{aligned} \int_\tau \partial_x \tilde{\lambda}_i \cdot 1 \, d\mu &= \text{Area} \cdot -\frac{1}{\text{Area}} \cdot y_{\vec{l}_i} = -y_{\vec{l}_i} \\ \int_\tau \partial_y \tilde{\lambda}_i \cdot 1 \, d\mu &= \text{Area} \cdot \frac{1}{\text{Area}} \cdot x_{\vec{l}_i} = x_{\vec{l}_i} \end{aligned}}$$

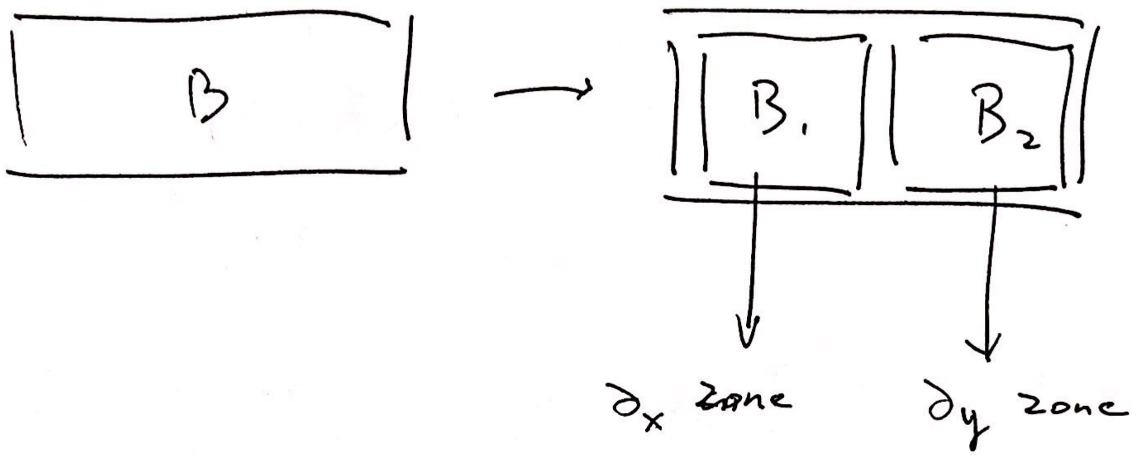
block B done.

$$\begin{array}{c} NE \\ NE \\ NT \end{array} \begin{bmatrix} A & O & \\ O & A & B^T \\ B & & O \end{bmatrix}$$

$$\sim \begin{bmatrix} \frac{1}{\text{Area}} \vec{l}_i \cdot \vec{l}_j & O & \\ O & \frac{1}{\text{Area}} \vec{l}_i \cdot \vec{l}_j & B^T \\ -y_{l_1} \; -y_{l_2} \; -y_{l_3} & x_{l_1} \; x_{l_2} \; x_{l_3} & O \end{bmatrix}$$

just check the index carefully

block B in details:



row: $2NE + t$ represents in t^{th} triangle / element.

this row (in B) ~~only~~ at most has 6 non-zero positions

they are $\partial_x u$ on 3 mid points

+

$\partial_y u$ on 3 mid points.

~~c~~ column: B_1: idx of edge

B_2: NE + idx of edge.

RHS subroutine:

$$\int f \tilde{\lambda}_1 \, d\mu$$

$$= \int f (1 - 2\lambda_1) \, d\mu$$

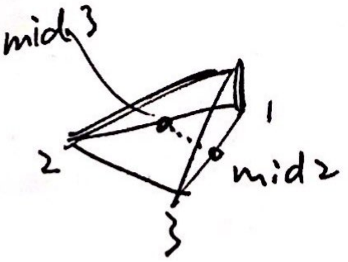
$$= \int f (\lambda_2 + \lambda_3 - \lambda_1) \, d\mu \quad \text{as } \lambda_1 + \lambda_2 + \lambda_3 = 1$$

$$= \int f \lambda_2 \, d\mu + \int f \lambda_3 \, d\mu - \int f \lambda_1 \, d\mu$$

solved in Poisson FEM RHS

used 3-points quadrature

to simulate integration.



$$H = \frac{f(\text{mid } 2) + f(\text{mid } 3)}{2}$$ instead of $f(1)$

to represent height.

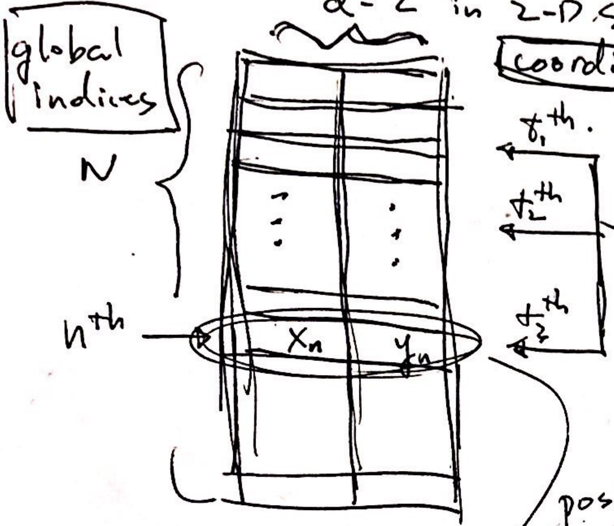
$$\int_{T} f \lambda_1 \, d\mu = \text{Volume} = \frac{1}{3} \cdot \text{Area} \cdot H$$

data structure

number of vertices (nodes)

node$(1:N,\ 1:d)$ dimension

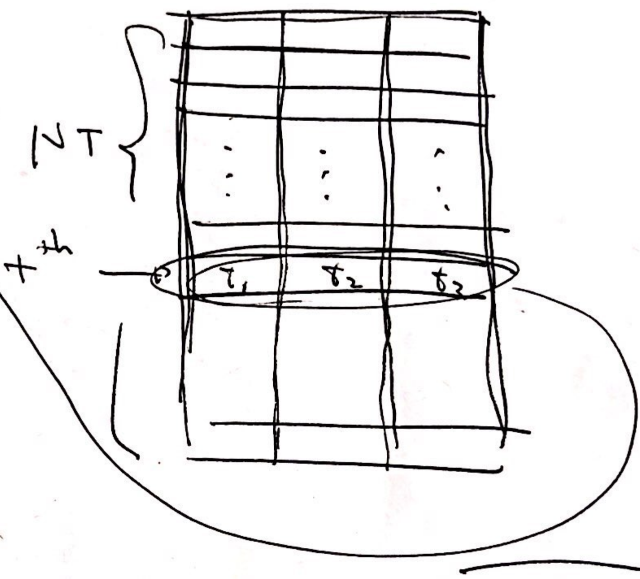
$d=2$ in 2-D space



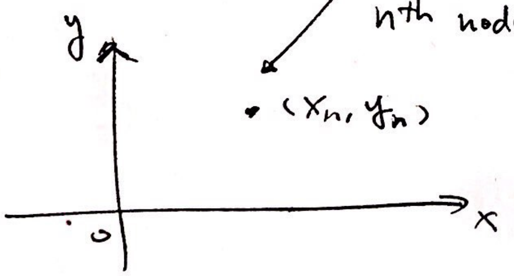
number of elements (triangles)

simplex in d-D space has d+1 vertices

elem$(1:NT,\ 1:d+1)$



position of n^{th} node

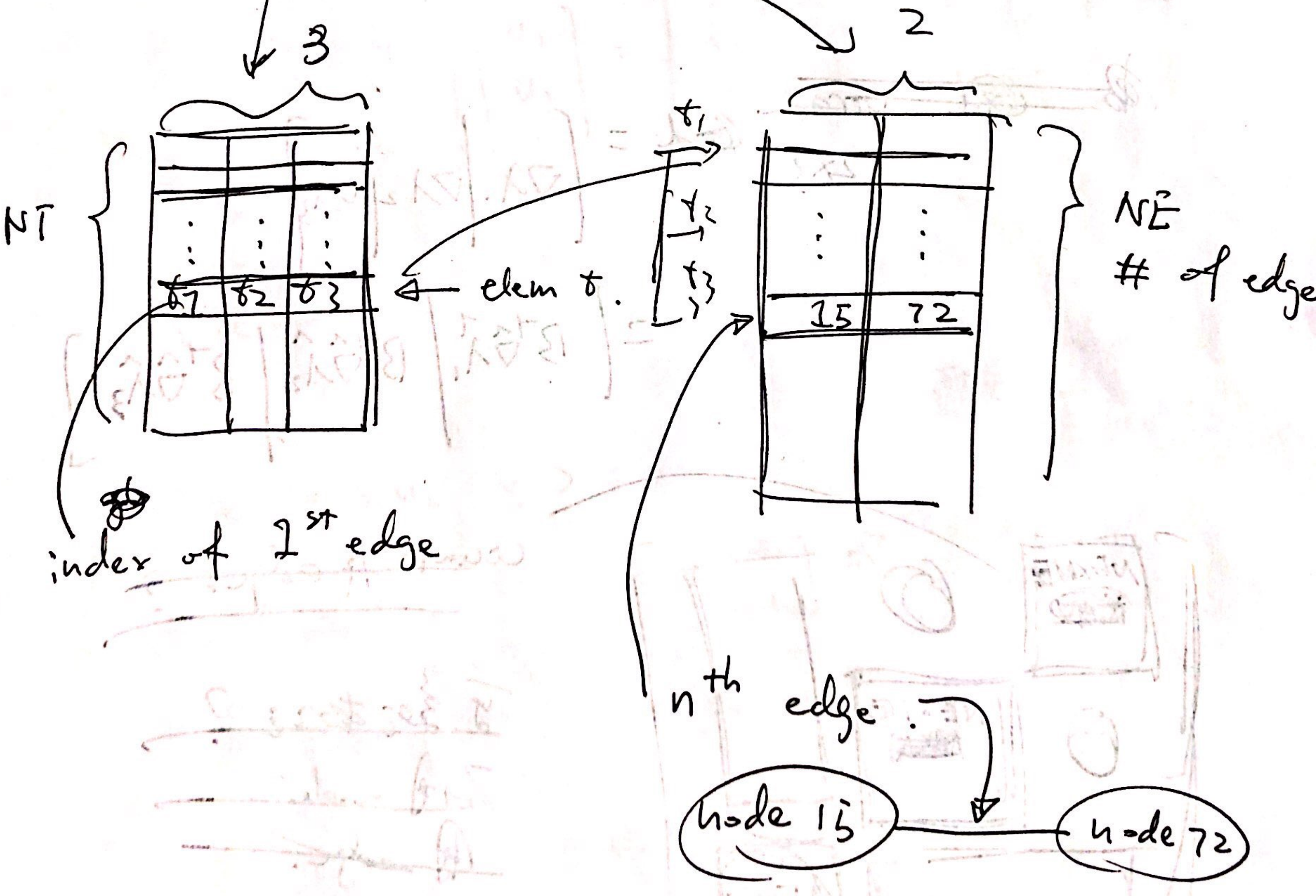


store $\boxed{\text{global indices}}$ of 3 vertices of given triangle. can further find their coordinates (positions) through "node"

node: global index $\xrightarrow[\text{to}]{\text{map}}$ coordinate

elem: element index $\xrightarrow[\text{to}]{\text{map}}$ global indices data structure.

[elem2edge, edge] = dofedge(elem)



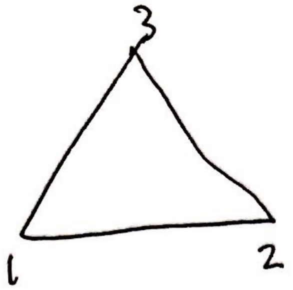
left < right
by global index of nod

$\tilde{\lambda}_1 = 1 - 2\lambda_1$

$\nabla\tilde{\lambda}_1 = -2\nabla\lambda_1$

$A := -\Delta$

$B := -\mathrm{div}$



$u_1 = c_1\tilde{\lambda}_1 + c_2\tilde{\lambda}_2 + c_3\tilde{\lambda}_3$

$u_2 = d_1\tilde{\lambda}_1 + d_2\tilde{\lambda}_2 + d_3\tilde{\lambda}_3$

$\int_T \partial_x u_1 \cdot 1 = \int_T \sum c_i \partial_x \tilde{\lambda}_i \cdot 1$

$$\begin{array}{cl} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} & u_1 \\ \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix} & u_2 \end{array}$$

At.

$$\left[\begin{array}{cc|c} A & 0 & B^T \\ 0 & A & \\ \hline B & & 0 \end{array}\right] \begin{bmatrix} u_1 \\ u_2 \\ p \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ g \end{bmatrix}$$

$B \uparrow [\partial_x \quad \partial_y]$

$p =$

identity.

$\vec{I}(x,y) = \begin{bmatrix} x \\ y \end{bmatrix}$

$(\partial_x u_1 + \partial_y u_2, q) = 0$

$\mathrm{div}\,\vec{u} = [1, 1] \cdot \begin{bmatrix} \partial_x u_1 \\ \partial_y u_2 \end{bmatrix} = \nabla\vec{I} \cdot \nabla\vec{u}$

$\langle \partial_x u_1, q \rangle + \langle \partial_y u_2, q \rangle = 0$

$(\partial_x \phi_i, \phi_j) + (\partial_y \phi_i, \phi_j) = 0$

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